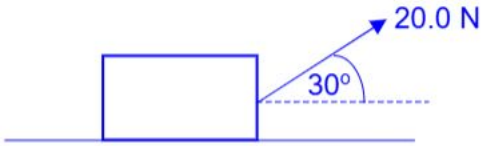
5 A man pulls a rope attached to a sled. The rope makes an angle of 30° with the horizontal. The man exerts a force of 20.0 N on the rope to move the sled horizontally through a distance of 40 m.

What is the work done by the man?

A 400 J **B** 490 J **C** 690 J **D** 800 J

Ans: C



Work done = (Force exerted by man in the direction of the motion of sled) x (distance moved)
= 20.0 cos30° x 40
= 692 J